

# Ali Siahkoohi

Assistant Professor  
Department of Computer Science  
University of Central Florida

alisk@ucf.edu  
<https://alishahkoohi.github.io>  
Last updated: July 16, 2026

## Research Interests

---

My research focuses on uncertainty quantification in complex systems that arise in computational science and engineering, with an emphasis on deep generative modeling. By integrating deep learning and scientific computing, my group develops scalable sampling techniques for high-dimensional distributions. This work enables Bayesian inference in large-scale PDE-based inverse problems (e.g., seismic and medical imaging) and provides a principled framework for building reliable, uncertainty-aware AI models for real-world applications.

## Academic Appointments

---

**University of Central Florida**  
Tenure-Track Assistant Professor  
Department of Computer Science

August 2025 – Present  
Orlando, FL, USA

**Rice University**  
Simons Postdoctoral Fellow  
Department of Computational Applied Mathematics & Operations Research  
Jointly hosted by Maarten V. de Hoop and Richard G. Baraniuk

August 2022 – July 2025  
Houston, TX, USA

## Education

---

**Georgia Institute of Technology**  
Ph.D. in Computational Science and Engineering  
Advised by Felix J. Herrmann

August 2022  
Atlanta, GA, USA

**University of Tehran**  
M.Sc. in Geophysics

March 2016  
Tehran, Iran

**Sharif University of Technology**  
B.Sc. in Electrical Engineering

August 2013  
Tehran, Iran

## Publications

---

Google Scholar profile: <https://scholar.google.com/citations?user=sxRMqYIAAAAJ&h>

### In Preparation & Under Review

- P5. [A. Siahkoohi](#). Amortized mean-shift interacting particles. Preprint arXiv:2606.15871, 2026  
[pdf] [code] [bib]
- P4. [A. Siahkoohi](#) and A. Thatipelli. A lift for input-convex neural network training. Preprint arXiv:2605.24274, 2026  
[pdf] [code] [bib]
- P3. A. Thatipelli, J. Sam, M. Louboutin, [A. Siahkoohi](#), R. Wang, and F. J. Herrmann. XConv: Low-memory stochastic backpropagation for convolutional layers. Preprint arXiv:2106.06998, 2026  
[pdf] [code] [bib]
- P2. [A. Siahkoohi](#), K. Aghazade, and A. Gholami. Dual-space posterior sampling for Bayesian inference in constrained inverse problems. Preprint arXiv:2603.00393, 2026a  
[pdf] [code] [poster] [bib]

- P1. A. Thatipelli and A. Siahkoochi. Hypernetwork-based approach for grid-independent functional data clustering. Preprint arXiv:2602.22823, 2026  
[pdf] [code] [poster] [bib]

## Journal Publications

- J9. A. Siahkoochi, R. Morel, R. Balestrieri, E. Allys, G. Sainton, T. Kawamura, and M. V. de Hoop. Multiscale clustering and source separation of InSight mission seismic data. *IEEE Transactions on Geoscience and Remote Sensing*, 64:1–16, 2026b  
[pdf] [code] [slides] [link] [bib]
- J8. R. Orozco, A. Siahkoochi, M. Louboutin, and F. J. Herrmann. ASPIRE: Iterative amortized posterior inference for Bayesian inverse problems. *Inverse Problems*, 41(4):045001, 2025  
[pdf] [code] [link] [bib]
- J7. R. Orozco, P. Witte, M. Louboutin, A. Siahkoochi, G. Rizzuti, B. Peters, and F. J. Herrmann. InvertibleNetworks.jl: A Julia package for scalable normalizing flows. *Journal of Open Source Software*, 9(99):6554, 2024  
[pdf] [code] [link] [bib]
- J6. L. Luzi, P. M. Mayer, J. Casco-Rodriguez, A. Siahkoochi, and R. G. Baraniuk. Boomerang: Local sampling on image manifolds using diffusion models. *Transactions on Machine Learning Research*, 2024a  
[pdf] [code] [link] [bib]
- J5. M. Louboutin, Z. Yin, R. Orozco, T. J. Grady II, A. Siahkoochi, G. Rizzuti, P. A. Witte, O. Møyner, G. J. Gorman, and F. J. Herrmann. Learned multiphysics inversion with differentiable programming and machine learning. *The Leading Edge*, 42(7):474–486, 2023  
[pdf] [link] [bib] [featured in Seismic Soundoff] [journal's most downloaded paper in '23]
- J4. Y. Zhang, Z. Yin, O. López, A. Siahkoochi, M. Louboutin, R. Kumar, and F. J. Herrmann. Optimized time-lapse acquisition design via spectral gap ratio minimization. *Geophysics*, 88(4):A19–A23, 2023a  
[pdf] [link] [bib]
- J3. A. Siahkoochi, G. Rizzuti, R. Orozco, and F. J. Herrmann. Reliable amortized variational inference with physics-based latent distribution correction. *Geophysics*, 88(3):R297–R322, 2023a  
[pdf] [slides] [code] [link] [bib] [featured in Geophysics Bright Spots]
- J2. A. Siahkoochi, G. Rizzuti, and F. J. Herrmann. Deep Bayesian inference for seismic imaging with tasks. *Geophysics*, 87(5):S281–S302, 2022a  
[pdf] [code] [link] [bib]
- J1. A. Siahkoochi, M. Louboutin, and F. J. Herrmann. The importance of transfer learning in seismic modeling and imaging. *Geophysics*, 84(6):A47–A52, 2019a  
[pdf] [code] [link] [bib]

## Peer-Reviewed Conference Proceedings

- C35. A. Siahkoochi and H. Oh. Conditional neural control variates for variance reduction in Bayesian inverse problems. In *Proceedings of the Forty-second Conference on Uncertainty in Artificial Intelligence*, 2026. In print  
[pdf] [code] [poster] [bib]
- C34. A. Siahkoochi and D. Sabeddu. On the role of memorization in learned priors for geophysical inverse problems. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, 2026. In print  
[pdf] [code] [slides] [bib]
- C33. K. Aghazade, A. Siahkoochi, and A. Gholami. Scalable Bayesian full waveform inversion via dual augmented Lagrangian SVGD. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, 2026. In print  
[pdf] [bib]

- C32. S. Alemohammad, J. Casco-Rodriguez, L. Luzi, A. I. Humayun, H. Babaei, D. LeJeune, A. Siahkoochi, and R. G. Baraniuk. Self-consuming generative models go MAD. In *The Twelfth International Conference on Learning Representations*, 2024  
[pdf] [extended pdf] [poster] [link] [bib] [featured in the news 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
- C31. L. Luzi, D. LeJeune, A. Siahkoochi, S. Alemohammad, V. Saragadam, H. Babaei, N. Liu, Z. Wang, and R. G. Baraniuk. Titan: Bringing the deep image prior to implicit representations. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 6165–6169, 2024b  
[pdf] [code] [link] [bib]
- C30. L. Baldassari, A. Siahkoochi, J. Garnier, K. Sølna, and M. V. de Hoop. Conditional score-based diffusion models for Bayesian inference in infinite dimensions. In *Advances in Neural Information Processing Systems*, volume 36, pages 24262–24290, 2023  
[pdf] [slides] [poster] [code] [link] [bib] [featured as a Spotlight presentation]
- C29. A. Siahkoochi, R. Morel, M. V. de Hoop, E. Allys, G. Sainton, and T. Kawamura. Unearthing InSights into Mars: Unsupervised source separation with limited data. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 31754–31772, 2023b  
[pdf] [slides] [poster] [code] [link] [bib]
- C28. R. Orozco, M. Louboutin, A. Siahkoochi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Amortized normalizing flows for transcranial ultrasound with uncertainty quantification. In *Medical Imaging with Deep Learning*, volume 227, pages 332–349, 2023a  
[pdf] [link] [bib]
- C27. R. Orozco, A. Siahkoochi, M. Louboutin, and F. J. Herrmann. Refining amortized posterior approximations using gradient-based summary statistics. In *5th Symposium on Advances in Approximate Bayesian Inference*, 2023b  
[pdf] [link] [bib]
- C26. R. Orozco, A. Siahkoochi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Adjoint operators enable fast and amortized machine learning based Bayesian uncertainty quantification. In *Medical Imaging 2023: Image Processing*, volume 12464, page 124641L, 2023c  
[pdf] [link] [bib]
- C25. Y. Zhang, Z. Yin, O. Lopez, A. Siahkoochi, M. Louboutin, and F. J. Herrmann. 3D seismic survey design by maximizing the spectral gap. In *Third International Meeting for Applied Geoscience & Energy*, 2023b  
[pdf] [poster] [bib]
- C24. A. Siahkoochi, M. Chinen, T. Denton, W. B. Kleijn, and J. Skoglund. Ultra-low-bitrate speech coding with pretrained Transformers. In *Proceedings of Interspeech*, pages 4421–4425, 2022b  
[pdf] [link] [bib]
- C23. A. Siahkoochi, M. Louboutin, and F. J. Herrmann. Velocity continuation with Fourier neural operators for accelerated uncertainty quantification. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1765–1769, 2022c  
[pdf] [slides] [code] [link] [bib]
- C22. M. Louboutin, P. Witte, A. Siahkoochi, G. Rizzuti, Z. Yin, R. Orozco, and F. J. Herrmann. Accelerating innovation with software abstractions for scalable computational geophysics. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1482–1486, 2022  
[pdf] [slides] [link] [bib]
- C21. Z. Yin, A. Siahkoochi, M. Louboutin, and F. J. Herrmann. Learned coupled inversion for carbon sequestration monitoring and forecasting with Fourier neural operators. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 467–472, 2022  
[pdf] [slides] [code] [link] [bib] [student oral paper honorable mention]
- C20. Y. Zhang, M. Louboutin, A. Siahkoochi, Z. Yin, R. Kumar, and F. J. Herrmann. A simulation-free seismic survey design by maximizing the spectral gap. In *Society of Exploration Geophysicists Technical*

- Program Expanded Abstracts*, pages 15–20, 2022  
[pdf] [slides] [code] [link] [bib]
- C19. [A. Siahkoohi](#), R. Orozco, G. Rizzuti, and F. J. Herrmann. Wave-equation based inversion with amortized variational Bayesian inference. In *EAGE Deep learning for seismic processing: Investigating the foundations workshop*, 2022d  
[pdf] [slides] [code] [link] [bib]
- C18. R. Orozco, [A. Siahkoohi](#), G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Photoacoustic imaging with conditional priors from normalizing flows. In *NeurIPS Workshop on Deep Learning and Inverse Problems*, 2021  
[pdf] [poster] [link] [bib]
- C17. [A. Siahkoohi](#), G. Rizzuti, M. Louboutin, P. Witte, and F. J. Herrmann. Preconditioned training of normalizing flows for variational inference in inverse problems. In *3rd Symposium on Advances in Approximate Bayesian Inference*, 2021  
[pdf] [slides] [code] [link] [bib]
- C16. [A. Siahkoohi](#) and F. J. Herrmann. Learning by example: Fast reliability-aware seismic imaging with normalizing flows. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1580–1585, 2021  
[pdf] [slides] [code] [link] [bib]
- C15. R. Kumar, M. Kotsi, [A. Siahkoohi](#), and A. Malcolm. Enabling uncertainty quantification for seismic data preprocessing using normalizing flows (NF)—An interpolation example. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1515–1519, 2021  
[pdf] [code] [link] [bib]
- C14. G. Rizzuti, [A. Siahkoohi](#), P. A. Witte, and F. J. Herrmann. Parameterizing uncertainty by deep invertible networks, an application to reservoir characterization. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1541–1545, 2020  
[pdf] [slides] [code] [link] [bib]
- C13. M. Zhang, [A. Siahkoohi](#), and F. J. Herrmann. Transfer learning in large-scale ocean bottom seismic wavefield reconstruction. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1666–1670, 2020  
[pdf] [slides] [code] [link] [bib]
- C12. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. Weak deep priors for seismic imaging. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2998–3002, 2020a  
[pdf] [slides] [code] [link] [bib]
- C11. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1636–1640, 2020b  
[pdf] [slides] [code] [link] [bib]
- C10. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2020c  
[pdf] [slides] [code] [link] [bib]
- C9. F. J. Herrmann, [A. Siahkoohi](#), and G. Rizzuti. Learned imaging with constraints and uncertainty quantification. In *NeurIPS Deep Inverse Workshop*, 2019  
[pdf] [slides] [poster] [link] [bib]
- C8. [A. Siahkoohi](#), R. Kumar, and F. J. Herrmann. Deep-learning based ocean bottom seismic wavefield recovery. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2232–2237, 2019b  
[pdf] [code] [slides] [link] [bib]

- C7. A. Siahkoohi, D. J. Verschuur, and F. J. Herrmann. Surface-related multiple elimination with deep learning. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 4629–4634, 2019c  
[pdf] [slides] [link] [bib]
- C6. G. Rizzuti, A. Siahkoohi, and F. J. Herrmann. Learned iterative solvers for the Helmholtz equation. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2019  
[pdf] [slides] [link] [bib]
- C5. A. Siahkoohi, M. Louboutin, R. Kumar, and F. J. Herrmann. Deep convolutional neural networks in prestack seismic—two exploratory examples. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2196–2200, 2018a  
[pdf] [poster] [link] [bib]
- C4. A. Siahkoohi, R. Kumar, and F. J. Herrmann. Seismic data reconstruction with generative adversarial networks. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2018b  
[pdf] [slides] [link] [bib]
- C3. A. Siahkoohi and A. Gholami. Sparsity promoting least squares migration for laterally inhomogeneous media. In *7th EAGE Saint Petersburg International Conference and Exhibition*, 2016  
[pdf] [link] [bib]
- C2. M. S. Ebrahimi, M. H. Daraei, J. Rezaei, and A. Siahkoohi. A novel utilization of wireless sensor networks as data acquisition system in smart grids. In *Materials Science and Information Technology*, volume 433-440, pages 6725–6730, 2012  
[pdf] [link] [bib]
- C1. A. Najafi, A. Siahkoohi, and M. B. Shamsollahi. A content-based digital image watermarking algorithm robust against JPEG compression. In *IEEE International Symposium on Signal Processing and Information Technology*, pages 432–437, 2011  
[pdf] [link] [bib]

## Theses

- T1. A. Siahkoohi. *Deep generative models for solving geophysical inverse problems*. PhD thesis, **Georgia Institute of Technology**, 2022  
[pdf] [slides] [link] [bib]

## Technical Reports

- R4. P. M. Mayer, L. Luzi, A. Siahkoohi, D. H. Johnson, and R. G. Baraniuk. Improving fairness and mitigating MADness in generative models. Technical Report arXiv:2405.13977, Rice University, 2024  
[pdf] [code] [slides] [bib]
- R3. L. Baldassari, A. Siahkoohi, J. Garnier, K. Sølna, and M. V. de Hoop. Taming score-based diffusion priors for infinite-dimensional nonlinear inverse problems. Technical Report arXiv:2405.15676, Rice University, 2024  
[pdf] [bib]
- R2. A. Siahkoohi, G. Rizzuti, P. A. Witte, and F. J. Herrmann. Faster uncertainty quantification for inverse problems with conditional normalizing flows. Technical Report arXiv:2007.07985, Georgia Institute of Technology, 2020d [pdf] [link] [bib]
- R1. A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Neural network augmented wave-equation simulation. Technical Report arXiv:1910.00925, Georgia Institute of Technology, 2019d  
[pdf] [code] [link] [bib]

## Awards

---

### Future Faculty Fellows Award

Rice University, George R. Brown School of Engineering and Computing  
[link]

June 2024  
Houston, TX, USA

## Selected Research Proposal Experience

---

### Scientific ML-supported subsurface characterization in physical function spaces

Awarded, 2024

- Funding Source: Occidental Petroleum Corporation, PI: Maarten V. de Hoop
- **Contributions:** Developed ideas and contributed to writing for two of the four research thrusts entitled “Score diffusion, nonlinear operators, and uncertainty quantification in function spaces” and “Unsupervised, factorial data decomposition and hidden signals: Reservoir characterization below salt, denoising, and monitoring”

### Learning and forecasting complex fault dynamics – Predictability of earthquakes

Not funded, 2024

- Funding Source: National Science Foundation, PI: Maarten V. de Hoop
- **Contributions:** Developed ideas and contributed to writing for one of the four research thrusts entitled “Structure in data, clustering, lattice theory, and diffusion models”

### Exploring the local geometry of deep networks

Awarded, 2023

- Funding Source: Office of Naval Research (DURIP), PI: Richard G. Baraniuk
- **Contributions:** Developed ideas and wrote research objectives for one of the three research thrusts entitled “The geometry of deep probabilistic models”

### A deep-learning framework for stable, interpretable, and uncertainty-quantified hybrid modeling of multi-scale complex systems

Not funded, 2023

- Funding Source: Department of Energy, PI: Pedram Hassanzadeh
- **Contributions:** Coordinated efforts within Richard G. Baraniuk’s group (a co-PI) to develop and write research objectives for one of the four research thrusts entitled “Spline operator-based analysis of Deep neural networks”

### Topological deep learning, causal inference, and data-driven forecasting for subsurface multiscale multiphysics systems

Awarded, 2022

- Funding Source: Department of Energy, PI: Maarten V. de Hoop
- **Contributions:** Led the effort to write the annual progress report

## Mentoring Experience

---

### University of Central Florida

Orlando, FL, USA

Departments of Computer Science & Electrical and Computer Engineering

- Banafsheh Adami [link] 2026 – Present  
CS Ph.D. Student — co-advised with Alain Kassab
- Anirudh Thatipelli [link] 2025 – Present  
CS Ph.D. Student — Ph.D. advisor
- Davide Sabeddu [link] 2025 – 2026  
ECE Ph.D. Student — Advised on the development of methods for mitigating overfitting in generative models and co-authored a paper (A. Siahkoohi and Sabeddu, 2026)

### Rice University

Houston, TX, USA

Department of Computational Applied Mathematics & Operations Research

- Jeffrey J. Sam [link] 2024 – 2025  
M.Sc. Student — Advised on the design and implementation of experiments and co-authored a paper (Thatipelli et al., 2026)
- Paul M. Mayer [link] 2022 – 2025

Ph.D. Student — Advised on the development of methods and software for two projects and co-authored two papers (Luzi et al., 2024a; Mayer et al., 2024)

### Georgia Institute of Technology

Atlanta, GA, USA

School of Computational Science and Engineering

- ▶ Rafael Orozco [\[link\]](#) 2020 – 2022  
Ph.D. Student — Advised on the development of methods and software for main Ph.D. thesis and co-authored four papers (Orozco et al., 2021, 2023b,c, 2025)
- ▶ Mi Zhang [\[link\]](#) 2019 – 2020  
Visiting Ph.D. Student (China University of Petroleum-Beijing) — Advised on the development of methods and software for a project and co-authored a paper (Zhang et al., 2020)

## Teaching Experience

---

### University of Central Florida

Orlando, FL, USA

Department of Computer Science

- ▶ Algorithms for Machine Learning [\[link\]](#) Fall 2026  
Instructor of Record
- ▶ Sampling Methods for Uncertainty Quantification [\[link\]](#) Spring 2026  
Instructor of Record
- ▶ Algorithms for Machine Learning [\[link\]](#) Fall 2025  
Instructor of Record

### Rice University

Houston, TX, USA

Department of Computational Applied Mathematics & Operations Research

- ▶ Numerical Analysis Fall 2024  
Substitute Instructor (12 lectures)
- ▶ Numerical Analysis I Fall 2022  
Substitute Instructor (18 lectures)

### Georgia Institute of Technology

Atlanta, GA, USA

School of Computational Science and Engineering

- ▶ Computational Foundations of Machine Learning Spring 2022  
Teaching Assistant
- ▶ Imaging with Data-Driven Models Fall 2019  
Teaching Assistant
- ▶ Numerical Analysis I Fall 2018  
Teaching Assistant

### Sharif University of Technology

Tehran, Iran

Department of Electrical Engineering

- ▶ Digital Signal Processing Spring 2011  
Teaching Assistant
- ▶ Signals and Systems Spring 2011  
Teaching Assistant
- ▶ Linear Algebra Spring 2010  
Teaching Assistant
- ▶ Electrical Engineering: Principles and Laboratory Fall 2009  
Teaching Assistant

## Talks

---

## Invited Talks

- |                                                                                                                                                                                                                                                                       |                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| T29. <b>IEEE Computer Society, Chapter of San Diego</b><br>Mitigating biases in self-consuming generative models<br>Open Research Institute (ORI)<br>[video]                                                                                                          | June 2025<br>Virtual oral presentation     |
| T28. <b>University of Central Florida</b><br>Towards reliable AI: A framework for quantification of AI uncertainty<br>Department of Computer Science                                                                                                                  | April 2025<br>Oral presentation            |
| T27. <b>Montana State University</b><br>Towards reliable AI: A framework for quantification of AI uncertainty<br>Gianforte School of Computing                                                                                                                        | March 2025<br>Oral presentation            |
| T26. <b>The University of California, Santa Barbara</b><br>Towards reliable AI: A framework for quantification of AI uncertainty<br>Department of Mechanical Engineering                                                                                              | February 2025<br>Oral presentation         |
| T25. <b>Johns Hopkins University</b><br>Towards reliable AI: A framework for quantification of AI uncertainty<br>Department of Electrical and Computer Engineering                                                                                                    | January 2025<br>Oral presentation          |
| T24. <b>ISCL Seminar Series, Pennsylvania State University</b><br>Mitigating biases in self-consuming generative models<br>Interdisciplinary Scientific Computing Laboratory (Dr. Romit Maulik)<br>[video]                                                            | November 2024<br>Virtual oral presentation |
| T23. <b>CNRS, Université Montpellier</b><br>Low-cost uncertainty quantification for large-scale inverse problems<br>RhEoVOLUTION Group (Dr. Andréa Tommasi)                                                                                                           | January 2023<br>Virtual oral presentation  |
| T22. <b>Workshop on Subsurface Uncertainty Description and Estimation</b><br>Reliable amortized variational inference with conditional normalizing flows via<br>physics-based latent distribution correction<br>International Meeting for Applied Geoscience & Energy | August 2022<br>Oral presentation           |
| T21. <b>Intelligent illumination of the Earth Workshop</b><br>Fast and reliability-aware seismic imaging with conditional normalizing flows<br>King Abdullah University of Science and Technology                                                                     | June 2021<br>Virtual oral presentation     |
| T20. <b>Advances in Seismic Imaging and Inversion Mini-symposium</b><br>Unsupervised data-guided uncertainty analysis in imaging and horizon<br>tracking<br>The 3rd Annual Meeting of the SIAM Texas–Louisiana Section                                                | October 2020<br>Virtual oral presentation  |

## Contributed Talks

- |                                                                                                                                                                  |                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| T20. <b>UCF Spring AI Research Day, UCF Institute of Artificial Intelligence</b><br>Multi-scale clustering and source separation of InSight mission seismic data | February 2026<br>Oral presentation |
| T19. <b>Geo-Mathematical Imaging Group Partners Meeting, Rice University</b><br>Improving fairness and mitigating MADness in generative models                   | November 2024<br>Oral presentation |
| T18. <b>International Conference on Machine Learning</b><br>Unearthing InSights into Mars: Unsupervised source separation with limited data                      | July 2023<br>Poster presentation   |
| T17. <b>Symposium on Advances in Approximate Bayesian Inference</b><br>Refining amortized posterior approximations using gradient-based summary<br>statistics    | July 2023<br>Poster presentation   |
| T16. <b>Geo-Mathematical Imaging Group Partners Meeting, Rice University</b><br>Martian time-series unraveled: A multi-scale nested approach with factorial      | May 2023<br>Oral presentation      |

variational autoencoders

- |      |                                                                                                                      |                                     |
|------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| T15. | <b>Geo-Mathematical Imaging Group Partners Meeting, Rice University</b>                                              | May 2023                            |
|      | Unearthing InSights into Mars: Unsupervised source separation with limited data                                      | Oral presentation                   |
| T14. | <b>International Meeting for Applied Geoscience &amp; Energy</b>                                                     | August 2022                         |
|      | Velocity continuation with Fourier neural operators for accelerated uncertainty quantification                       | Oral presentation                   |
| T13. | <b>Chrome Media Team, Google</b>                                                                                     | December 2021                       |
|      | Low-bitrate speech coding with Transformers                                                                          | Virtual oral presentation           |
| T12. | <b>ML4SEISMIC Partners Meeting, Georgia Institute of Technology</b>                                                  | November 2021                       |
|      | Multifidelity conditional normalizing flows for physics-guided Bayesian inference                                    | Virtual oral presentation           |
| T11. | <b>ML4SEISMIC Partners Meeting, Georgia Institute of Technology</b>                                                  | November 2021                       |
|      | Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach            | Virtual oral presentation           |
| T10. | <b>Society of Exploration Geophysicists International Exposition and Annual Meeting</b>                              | September 2021                      |
|      | Learning by example: Fast reliability-aware seismic imaging with normalizing flows<br>[video]                        | Virtual oral presentation           |
| T9.  | <b>Symposium on Advances in Approximate Bayesian Inference</b>                                                       | January 2021                        |
|      | Preconditioned training of normalizing flows for variational inference in inverse problems<br>[video]                | Prerecorded short oral presentation |
| T8.  | <b>European Association of Geoscientists &amp; Engineers Annual Conference &amp; Exhibition</b>                      | December 2020                       |
|      | A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification                            | Virtual oral presentation           |
| T7.  | <b>Society of Exploration Geophysicists International Exposition and Annual Meeting</b>                              | October 2020                        |
|      | Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach<br>[video] | Virtual oral presentation           |
| T6.  | <b>Society of Exploration Geophysicists International Exposition and Annual Meeting</b>                              | October 2020                        |
|      | Weak deep priors for seismic imaging<br>[video]                                                                      | Virtual oral presentation           |
| T5.  | <b>Society of Exploration Geophysicists Student Chapter, Georgia Tech</b>                                            | February 2020                       |
|      | A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification                            | Oral presentation                   |
| T4.  | <b>HotCSE Seminar, CSE Department, Georgia Institute of Technology</b>                                               | November 2019                       |
|      | Learned imaging with constraints and uncertainty quantification                                                      | Oral presentation                   |
| T3.  | <b>Society of Exploration Geophysicists International Exposition &amp; Annual Meeting</b>                            | September 2019                      |
|      | Deep-learning based ocean bottom seismic wavefield recovery                                                          | Oral presentation                   |
| T2.  | <b>Society of Exploration Geophysicists International Exposition &amp; Annual Meeting</b>                            | September 2019                      |
|      | Surface-related multiple elimination with deep learning                                                              | Oral presentation                   |
| T1.  | <b>Society of Exploration Geophysicists International Exposition &amp; Annual Meeting</b>                            | October 2018                        |
|      | Deep convolutional neural networks in prestack seismic—two exploratory examples                                      | Poster presentation                 |

## Professional Service

---

## Editorial Service

- ▶ **Acta Geophysica**, Associate Editor 2024 – Present  
Applied Geophysics section
- ▶ **Journal of Mathematics**, Guest Editor 2023 – 2024  
Special issue on Applied Mathematics in Inverse Problems and Uncertainty Quantification

## Conference Organization

- ▶ **International Meeting for Applied Geoscience & Energy**, Session Chair 2022

## Technical Program Committee Member and Reviewer

- ▶ The Conference on Uncertainty in Artificial Intelligence (UAI) 2026
- ▶ Symposium on Probabilistic Machine Learning (ProbML) 2026
- ▶ International Conference on Machine Learning (ICML) 2024 – 2026
- ▶ International Conference on Learning Representations (ICLR) 2024 – 2026
- ▶ Annual AAAI Conference on Artificial Intelligence 2025 – 2026
- ▶ Structured Probabilistic Inference & Generative Modeling 2023 – 2025
- ▶ Frontiers in Probabilistic Inference: Sampling Meets Learning 2025
- ▶ Neural Information Processing Systems (NeurIPS) 2023 – 2025
- ▶ Artificial Intelligence and Statistics Conference (AISTATS) 2024 – 2025
- ▶ Advances in Approximate Bayesian Inference (AABI) 2023 – 2024
- ▶ Structured Probabilistic Inference & Generative Modeling (ICML workshop) 2023 – 2024
- ▶ International Speech Communication Association (Interspeech) 2023
- ▶ Deep Generative Models for Health (NeurIPS workshop) 2023
- ▶ International Meeting for Applied Geoscience & Energy 2023

## Journal Reviewer

- ▶ Transactions on Machine Learning Research
- ▶ IEEE Transactions on Neural Networks and Learning Systems
- ▶ IEEE Transactions on Computational Imaging
- ▶ Journal of Medical Imaging
- ▶ SIAM Journal on Scientific Computing
- ▶ Notices of the American Mathematical Society (AMS)
- ▶ Entropy
- ▶ IEEE Transactions on Geoscience and Remote Sensing
- ▶ IEEE Geoscience and Remote Sensing Letters
- ▶ Remote Sensing
- ▶ Geosciences
- ▶ Geophysics
- ▶ The Leading Edge
- ▶ Geophysical Prospecting
- ▶ Journal of Geophysical Research – Solid Earth

## Industry Research Experience

### Google

Research Intern (cf. [A. Siahkoohi et al. \(2022b\)](#))  
Chrome Media Team

August 2021 – December 2021  
San Francisco, CA, USA

## Selected Media Coverage

---

|                                                                                                                                                                             |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Future Faculty Fellow Ali Siahkoohi joins University of Central Florida as assistant professor</b><br>Rice University Engineering News<br><a href="#">[link]</a>         | June 2025     |
| <b>AI's Mad Loops</b><br>Rice Magazine<br><a href="#">[link]</a>                                                                                                            | February 2025 |
| <b>AI Appears to Be Slowly Killing Itself</b><br>Futurism<br><a href="#">[link]</a>                                                                                         | August 2024   |
| <b>When A.I.'s Output Is a Threat to A.I. Itself</b><br>The New York Times<br><a href="#">[link]</a>                                                                        | August 2024   |
| <b>Breaking MAD: Generative AI could break the internet</b><br>Rice News, Rice University<br><a href="#">[link]</a>                                                         | July 2024     |
| <b>'Cesspool of AI crap' or smash hit? LinkedIn's AI-powered collaborative articles offer a sobering peek at the future of content</b><br>Fortune<br><a href="#">[link]</a> | April 2024    |
| <b>AI's 'mad cow disease' problem tramples into earnings season</b><br>Yahoo!finance<br><a href="#">[link]</a>                                                              | April 2024    |
| <b>'Mad' AI risks destroying the Information Age</b><br>The Telegraph<br><a href="#">[link]</a>                                                                             | February 2024 |
| <b>When AI Is Trained on AI-Generated Data, Strange Things Start to Happen</b><br>Futurism<br><a href="#">[link]</a>                                                        | August 2023   |
| <b>Episode 194: Improving integration in machine learning workflows</b><br>Seismic Soundoff Podcast, Society of Exploration Geophysicists<br><a href="#">[link]</a>         | July 2023     |
| <b>Training AI With Outputs of Generative AI Is Mad</b><br>CDOtrends<br><a href="#">[link]</a>                                                                              | July 2023     |
| <b>AI's trained on AI-generated images produce glitches and blurs</b><br>NewScientist<br><a href="#">[link]</a>                                                             | July 2023     |
| <b>Scientists make AI go crazy by feeding it AI-generated content</b><br>TweakTown<br><a href="#">[link]</a>                                                                | July 2023     |
| <b>AI Loses Its Mind After Being Trained on AI-Generated Data</b><br>Futurism<br><a href="#">[link]</a>                                                                     | July 2023     |
| <b>Generative AI Goes 'MAD' When Trained on AI-Created Data Over Five Times</b><br>Tom's Hardware<br><a href="#">[link]</a>                                                 | July 2023     |

**Group Brings Seismic Imaging to Climate-Change Conversations and Beyond**  
College of Computing News, Georgia Institute of Technology  
[\[link\]](#)

August 2022